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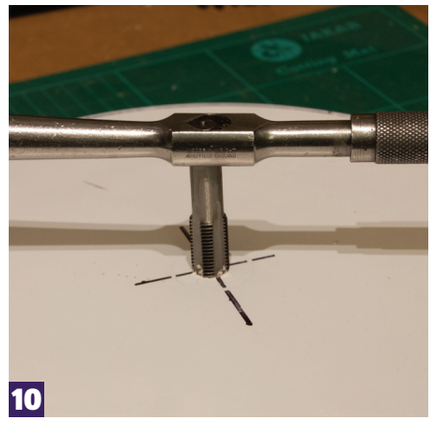
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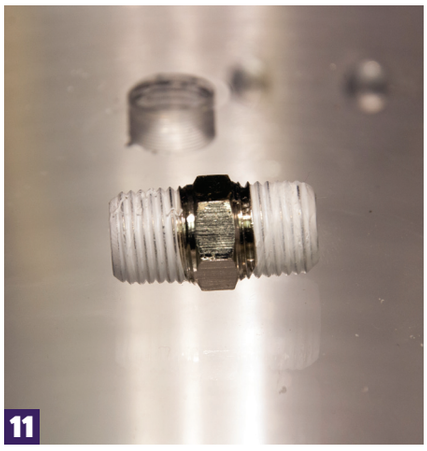
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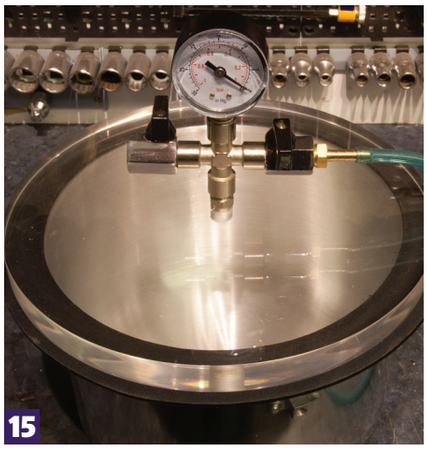
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16

good quality abrasive will do. The sharp edges can be rounded off too at this stage

9 You could leave the edge with a matte finish from the sanding, but I decided to go for a more professional polished look so brought the Perspex to a clear, glass like finish, with Farecla 300 and 500 polishing compounds

10 Remove the workpiece from the lathe and return it to the pillar drill to be drilled out to 11.8mm and then tapped by hand using a ¼in BSP tap in a suitable tap wrench

11 The male threads of all the components are wound with a few turns of plumber's PTFE tape to ensure good airtight joints. Wrap the tape in the direction of the thread

12 Begin the assembly by threading the straight connector into the tapped hole in the Perspex lid. Tighten it with a spanner sufficiently to ensure an airtight seal

13 The remaining components are then assembled as shown in the photograph. I found it easiest to assemble the components to the cross piece first and then attach the completed assembly to the connector already fitted to the Perspex lid. The lid is then fitted using the original seal from the pressure cooker as a gasket

14 Connect the vacuum supply and run up to its maximum to test the completed chamber. Then close the inlet valve, turn off the air supply to the vacuum generator and leave the assembly to stand to see if the vacuum holds up

15 After 24 hours the vacuum in my chamber had dropped from 25inHg to about 10inHg so in order to achieve a better seal I replaced the original seal with a flat rubber sealing washer of the type used for industrial pipework. These are readily available in a wide range of sizes from specialist suppliers. You can omit this step if the original seal on your chamber is good enough

16 All that now remains is a trial run with some punky timber to test the equipment. I decided to use a couple of pen blanks that had been cut from some soft horse chestnut (*Aesculus hippocastanum*), which was quite crumbly to turn

HANDY HINTS

1. Always read any instructions when using equipment. High pressure and chemicals make for a potentially nasty mix if things go wrong. Double check everything too.
2. Softer timbers, not dense close-grained ones – will yield the best stabilising results.

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